SMART IOT- based Pet Feeder

Week 2 progress

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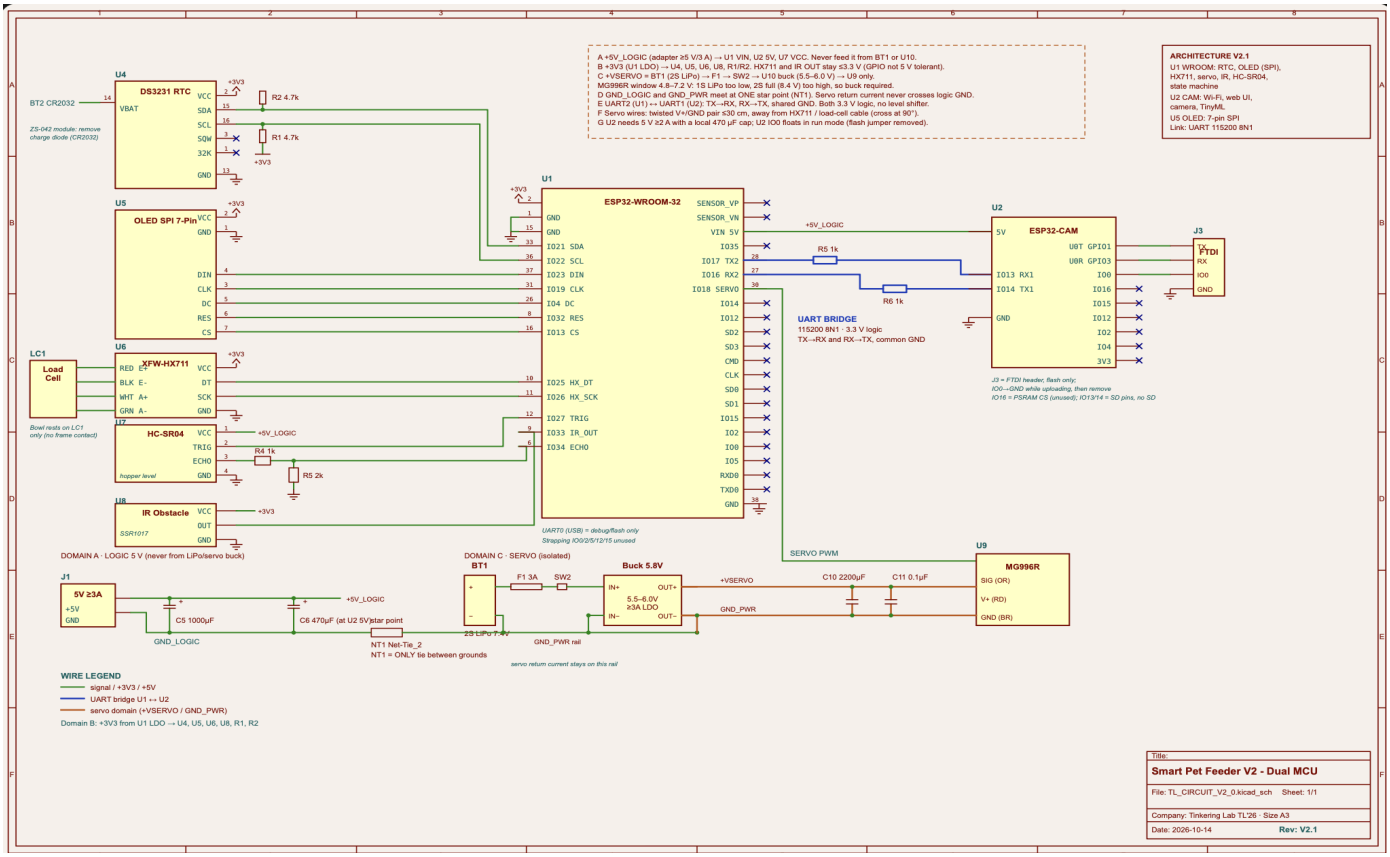
1 .3D MODEL



3D visualization :
https://drive.google.com/file/d/14zNzjED0UQP0isRSEqo19jVAh3GhFKmZ/view?usp=drive_link

The 3D model features a modular, print-ready housing for the Smart Pet Feeder. The primary structure consists of a cuboidal hopper mounted to the base, designed with a single-sided internal slope to ensure consistent, gravity-fed dispensing. At the bottom of this slope, there is a mechanical gate operated by an MG996R servo motor. An angled chute extends outward from the gate to the feeding bowl, maintaining a comfortable physical clearance so the main hopper body does not interfere with the pet while eating. Directly beneath the bowl lies an isolated mounting pedestal for the load cell, allowing the bowl to float independently for accurate HX711 weight measurements. Finally, a dedicated enclosure at the rear of the base houses the ESP microcontrollers, featuring internal channels that route all sensor and motor wiring safely away from the food path.

2. Circuit Diagram

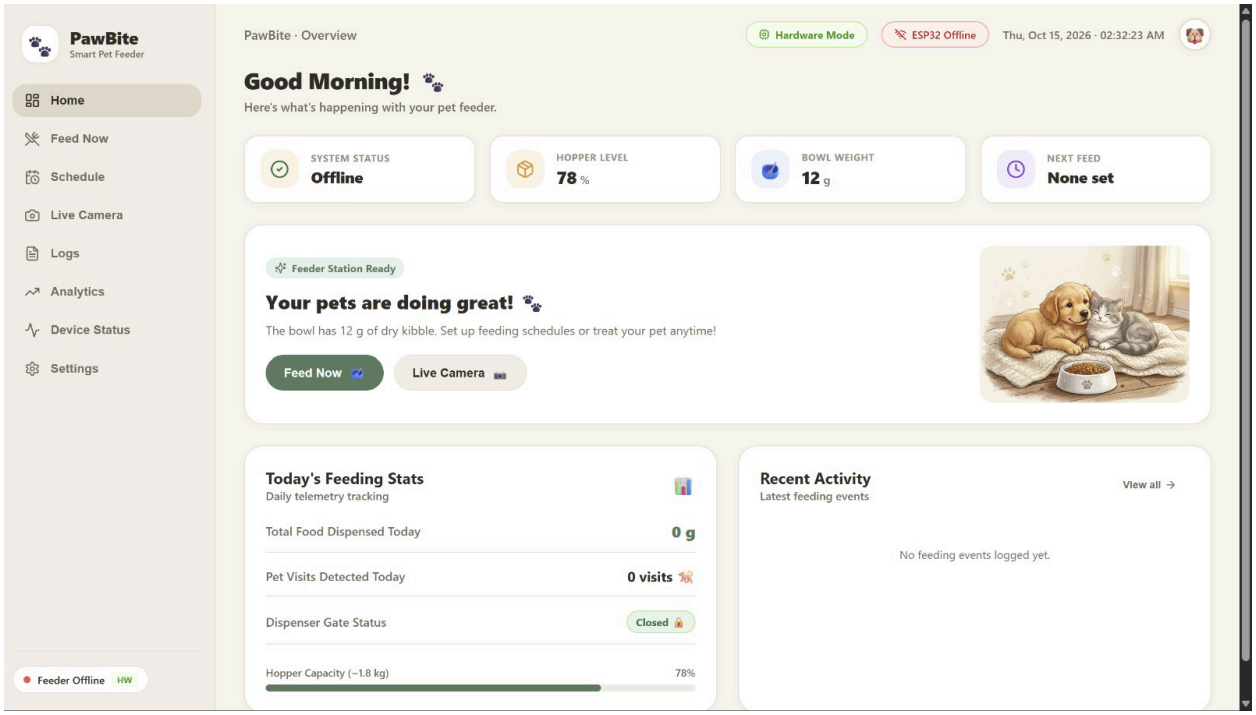


The electronic schematic for the Smart Pet Feeder features a robust Dual-MCU architecture designed in KiCad. The system is divided into two main processing units communicating via a protected UART bridge: an ESP32-WROOM-32 dedicated to real-time sensor polling and motor control, and an ESP32-CAM handling web connectivity. The main ESP32-WROOM-32 directly interfaces with a 7-pin SPI OLED display, a DS3231 RTC, an XFW-HX711 load cell amplifier, an HC-SR04 ultrasonic sensor (utilizing a 1k/2k voltage divider for logic protection), and an IR obstacle sensor. To guarantee system stability and prevent brownouts, the design implements strict power isolation. A 5V main adapter supplies the logic domain, while a completely isolated 2S LiPo battery 7.4V and 5.8V buck converter specifically power the high-current MG996R servo motor. These two power domains are joined only at a single star ground point to prevent motor noise from interfering with the sensitive weighing electronics and microcontrollers.

3. IOT - Platform

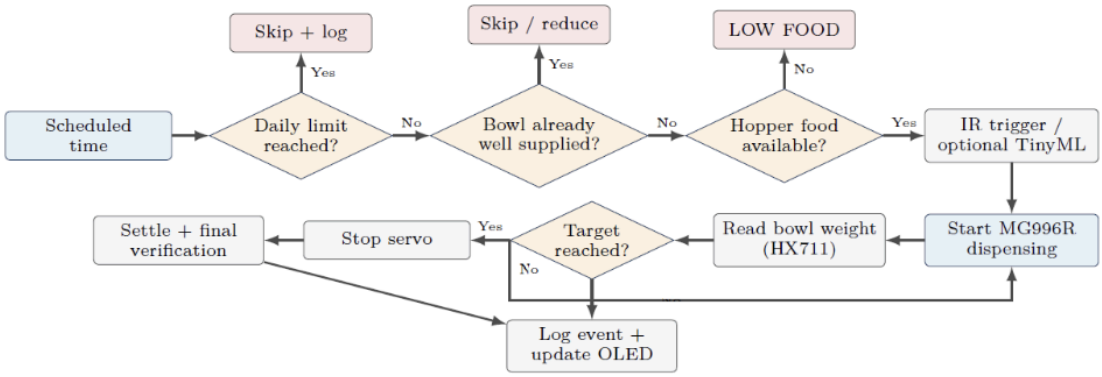
The Smart Pet Feeder V2.0 features a React/TypeScript web dashboard that communicates via HTTP and SSE to a dual-MCU hardware architecture. An ESP32-CAM handles networking and

camera interfaces, while an ESP32-WROOM manages real-time sensing, load cell monitoring, and servo actuation via a UART bridge. While the frontend software builds, mock device services, and simulated protocol tests have passed successfully, the end-to-end hardware integration remains incomplete. Currently, the WROOM firmware lacks handlers for scheduling, calibration, and full telemetry, and the CAM does not reliably await command acknowledgments. Because no physical hardware tests or firmware builds have been fully verified yet, future work must resolve these open protocol incompatibilities to validate actual CAM-to-WROOM communication and physical feeding.



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4. ESP 32 CODE



The Smart Pet Feeder uses an ESP32-WROOM-32D as its main sensor and actuator controller. It interfaces with a load cell through an HX711 amplifier to measure dispensed food, a servo motor to operate the dispensing gate, an ultrasonic sensor to estimate hopper food levels, an IR sensor to detect pet presence, and a DS3231 RTC for timekeeping. An OLED provides local status information, while the Preferences library stores feeding data in non-volatile memory.

The controller uses a state-machine-based feeding process that includes pre-checks, optional AI-based pet verification, weight-monitored dispensing, settling, top-up correction and reporting. It communicates with a separate ESP32-CAM over UART (Serial2). The planned CAM unit handles camera-based pet verification and web-dashboard interaction, while the main ESP32 remains responsible for the physical feeding mechanism. The current firmware provides the core control structure and communication protocol, with scheduled feeding and some recovery and configuration features still requiring implementation.

5. CHALLENGES FACED

1. **Circuit Design and Power Management:** Planning component connections, voltage compatibility, grounding, and a stable power supply for the ESP32, sensors, and servo motor.
2. **Firmware Development:** Structuring the ESP32 code to handle multiple sensors, RTC operations, weight monitoring, and servo control efficiently.
3. **Accurate Food Dispensing:** Planning load-cell-based weight measurement and accounting for food overshoot and possible blockage in the dispensing mechanism.
4. **UART Communication:** Defining a reliable communication protocol between the main ESP32 and ESP32-CAM for feeding commands, pet verification, and status updates.

5. **IoT Platform Integration:** Coordinating dashboard functions with the embedded system for remote feeding control and real-time status monitoring.
6. **Mechanical Design:** Designing the 3D model to ensure proper food flow, stable construction, and suitable placement of the servo, bowl, sensors, and electronics.
7. **System Integration:** Ensuring compatibility between the firmware, circuit diagram, IoT platform, and mechanical design before proceeding to complete hardware testing.

6.WORKLOAD DISTRIBUTION

Contributions	Topics	Details
Harsh Thapar **Keerat Singh**	ESP32 and Code	ESP32 firmware, state machine, scheduling, RTC interfacing, schedule persistence, system integration
Gurnoor Singh **Gorish Lamba**	IOT Platform	Wi-Fi web interface and feeding log, fault and test cases, performance evaluation, documentation and presentation
Gorish Lamba **Chayan Juneja**	Mechanical Design	Food container, servo-operated dispensing gate, bowl and load-cell mounting, jam prevention
Harsh Thapar **Keerat Singh** **Chayan Juneja** **Gurnoor Singh**	Circuit Design	Circuit design and wiring, power supply and servo isolation, IR sensor integration, display and buttons interfacing